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PERFORMANCE SPECIFICATION

ENERGY-ATTENUATING FLOOR MATS FOR USE IN U.S. MILITARY GROUND VEHICLES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Comments, suggestions, or questions on this document should be addressed to U.S. Army CCDC, Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-IES, MS #267, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or send email to usarmy.detroit.ccdc-gvsc.mbx.standardization@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1 SCOPE

1.1 Scope. This specification establishes performance and operating requirements for Energy-Attenuation (EA) floor mats used for all U.S. Army military vehicles and reporting procedures to be used for requirements verification. EA floor mats reduce the risk of injury to occupants during mine blast accelerative loading events. This document should be used in conjunction with the vehicle program specific performance specification which entails requirements tailored to the intended vehicle specific design, protection location(s) and associated geometry. Unless otherwise specified by the Government, the EA floor mat (as defined in section 6.3(b)) should ensure threshold requirements are met (as defined in section 6.3(e)).

1.2 Application. Certain text found in this document is generic in nature but further elaboration can be found in section 6.3 or in the vehicle specified contract.

2 APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810	Environmental Engineering and Laboratory Tests
MIL-STD-1472	Human Engineering Design Guidelines
MIL-STD-3010	Test Procedures for Packaging Materials and Containers

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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CODE OF FEDERAL REGULATIONS

40 CFR 82	-	Environmental Protection Agency; Protection of Stratospheric Ozone
49 CFR 572	-	Anthropomorphic Test Devices

(Copies of these documents are available online at https://www.ecfr.gov.)

ARMY PUBLIC HEALTH CENTER'S TOXICITY EVALUATION PROGRAM

AR 40-5	-	Preventive Medicine
DA PAM 70-3	-	Army Acquisition Procedures

(Copies of these documents are available online at https://www.apd.army.mil.)

U.S. ARMY COMBAT CAPABILITIES DEVELOPMENT COMMAND (CCDC) GROUND VEHICLE SYSTEMS CENTER (GVSC) TECHNICAL REPORTS (TR)

TR 27790	-	OCP TECD Enhanced Injury Assessment Reference Values
TR 27709	-	Laboratory Test Procedure for Occupant Protection Lab Subsystem Drop Tower (LQP-25-WI07, LQP-25-WI08)

(Copies of these documents are available online at https://www.dtic.mil.)

U.S. ARMY COMBAT CAPABILITIES DEVELOPMENT COMMAND (CCDC) GROUND VEHICLE SYSTEMS CENTER (GVSC) LABORATORY PROCEDURES

LQP-25-04	-	Floor Interface Technology Accelerator Test Procedure
LQP-25-WI07	-	Anthropomorphic Test Device (ATD) Setup and Positioning Work Instruction
LQP-25-WI08	-	High Speed Camera Setup and Operation Work Instruction
LQP-25-WI09	-	Anthropomorphic Test Device (ATD) Lower Body Fixture Setup and Positioning Work Instruction

(Copies of these documents are available by contacting the Occupant Protection Lab Manager at 44030 Schilling Street, Bldg 1426, Harrison Twp., MI 48045)

NATIONAL HIGHWAY TRAFFIC SAFETY ADMINISTRATION (NHTSA)

TP 208-14 Section 12.2	-	Test Temperature Conditions

(Copy of this document is available online at https://www.nhtsa.gov/vehicle-manufacturers/test-procedures.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise indicated, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NFSI B101.1	-	Test Method for Measuring Wet Static Coefficient of Friction of Common Hard-Surface Floor Materials

(A copy of this document is available online at https://www.ansi.org.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) INTERNATIONAL

ASTM D2240	-	Standard Test Method for Rubber Property - Durometer Hardness
ASTM E1354	-	Rates for Materials and Products Using an Oxygen Consumption Calorimeter

(Copies of these documents are available online at https://www.astm.org)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. If a conflict exists between a metric measurement and an English (Imperial) measurement, the metric measurement should take precedence. If there is a conflict between references, the contractor is required to contact, in writing, the contracting activity for clarification. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

(Section 3 requirements are defined as threshold requirements that provide a start point and general reference to extract and begin the requirements tailoring process to meet the specific fit, form, function needs of the EA floor mat(s) to be purchased by the acquisition team. It is expected that the acquisition team shall tailor Section 3 requirements in the contract and shall build upon the Section 3 minimum threshold requirements as required by the acquisition program.)

3.1 First article inspection and lot acceptance tests (LATs). First article inspection shall be performed on one complete energy-attenuation (EA) floor mat when a first article sample is required. The completed EA floor mat shall conform to Section 3 requirements and the verification methods as listed in Table I. The contract shall specify procedures for conducting first article inspection of sample items and requirements for conducting LATs (see 4.1).

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3.2 Materials. The contractor shall select materials capable of meeting all of the operational and environmental requirements contained herein and as stated in the contract (see 4.2).

3.3 General.

3.3.1 Documentation. Documentation of requirements conformance results shall be retained in the contractor's files in auditable condition and submitted to the purchasing agency (PA) upon request (see 4.3.1).

3.3.2 Inspection conditions. Unless otherwise specified in the contract or in a following requirement section, all inspections shall be conducted under the following ambient environmental conditions (see 4.3.2):

Temperature:	20 ±15°C (68 ±27°F)
Barometric pressure:	Facility ambient
Humidity:	Up to 95% relative humidity

3.3.3 Units of measure. Metric units with U.S. Standard units expressed in parenthesis after metric units (i.e., 2.54 cm (1.00 in.)) shall be used (see 4.3.3).

(Exceptions to this requirement shall include documentation required in 3.4 "Survivability" of this specification and unless otherwise specified in the contract.)

3.3.4 Different thicknesses. When the contract calls for floor mats of different thicknesses, each thickness must meet the requirements herein (see 4.3.4).

3.4 Survivability.

3.4.1 Test fixture and input. The EA floor mat shall meet Hybrid III lower half (50th percentile male ATD (T), 5th percentile female and 95th percentile male ATD (O)) injury criteria in requirement 3.4.4 using the Floor Interface Technology Accelerator (FITA) (see 4.4.1).

3.4.2 Anthropomorphic test device (ATD). The Hybrid III shall be used as the lower leg surrogate utilizing the lower half ATD with the GVSC lower body fixture (see 4.4.2).

3.4.3 Further instrumentation. At least two accelerometers at opposite edges shall be mounted on the floating plate, temperature of the floor mat, boots and tibia flesh shall be measured with an infrared thermometer prior to each test and all tests shall be recorded with high-speed video cameras with at least 2 views, one overall view and one close-up at the boots. If possible, the use of a mirror is recommended to gain additional perspectives (see 4.4.3).

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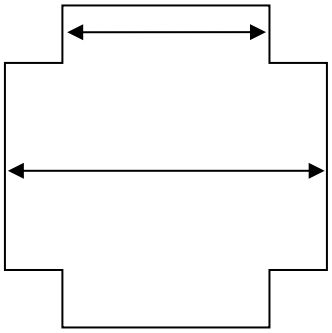
3.4.4 Injury criteria. The SI (metric) System of units shall be used for Anthropomorphic Test Device (ATD) test data to assess injury responses for the 50th percentile male ATDs. These Enhanced Injury Assessment Reference Values shall be utilized for the body regions listed below (see 4.4.4):

a. Upper Tibia Compressive Force, Z-direction (CFC 600)
b. Lower Tibia Compressive Force, Z-direction (CFC 600)

(For supplemental information on injury values, please refer to Section 6.4.)

3.4.5 Test conditions. Test for temperature within 36 inches of the lower half ATD before test impact. Alternately, the ATD outside surface temperature of one knee can be taken to verify the surface temperature before to test impact (see 4.4.5).

3.4.6 Test specimen size. The floor mat test sample shall be at least 38.1 x 38.1 centimeters (15 x 15 inches) to provide space for both boots. It shall fit within the following shape (Figure 1) (see 4.4.6).



a = 419.1 millimeters (16.5 inches)
b = 609.6 millimeters (24 inches)
Double axisymmetric

Figure 1: Boundary shape for floor mat test samples.

3.5 Environmental considerations. The use of toxic chemicals or ozone-depleting chemicals shall be prohibited (see 4.5).

3.6 AIPH TEP toxicity clearance. A toxicity clearance shall be obtained from the Army Public Health Center (APHC), Toxicology Portfolio, Toxicity Evaluation Program (TEP) in accordance with AR 40-5 and DA PAM 70-3 (see 4.6).

3.7 Wet surface traction. The walking surface wet static coefficient of friction shall be 0.50 or greater (see 4.7).

3.8 Self-draining water immersion. After water immersion the mat shall be self-draining. The material shall not retain more than 10% of its weight on, below or within it after water immersion. Exposure to water shall have no adverse material effects and shall not hinder EA of section 3.4.4 (see 4.8).

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3.9 High storage temperature. Exposure to high storage temperatures shall have no adverse material effects and shall not hinder EA of section 3.4.4 (see 4.9).

3.10 Low storage temperature. Exposure to low storage temperatures shall have no adverse material effects and shall not hinder EA of section 3.4.4 (see 4.10).

3.11 Heat release rate (fire resistance).

3.11.1 Average heat release rate. Unless otherwise specified in the contract, with the exception of fabrics and textiles, the EA floor mat, shall be tested and evaluated. No melting, dripping or pooling shall be permitted (see 4.11.1).

3.11.2 Average peak heat release rate. Unless otherwise specified in the contract, with the exception of fabrics and textiles, the Average Peak Heat Release Rate of a minimum of 3 samples shall be less than 85 kW/m^2 (see 4.11.2).

3.12 Fungal susceptibility. The material used to construct the EA floor mat shall be fungi inert or the material fungi contamination rating shall be 1 or less (if not otherwise specified in the contract) for a twenty- eight (28) day minimum test period, unless otherwise specified in the contract (see 4.12).

3.13 Fluid contamination. The material used to construct the EA floor mat shall be compatible with intermittent contamination of contaminant fluid groups and test fluids (see 4.13).

3.14 Integration requirements.

3.14.1 Mounting. The EA floor mat shall be mountable (see 4.14.1).

3.14.1.1 Adhesive bonding. Direct glue-down or permanent structural adhesive bonding methods shall not be used (see 4.14.1.1).

3.14.1.2 Tools. Mounting provisions shall use standard tools (no special tools) used in the General Mechanics Tool Kit for the purpose of installation, maintenance, service, and removal. Mounting provisions shall implement safeguards and not become detached from the attachment points or become secondary projectiles when subjected to the range of vibration, acceleration, shock, blast, and other impact tests specified in the contract. If the joint at the attachment point is completely severed, the attachment point shall be considered detached (see 4.14.1.2).

3.14.2 Maintenance. The EA floor mat shall be maintainable according to handling requirements in MIL-STD-1472 (see 4.14.2).

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3.14.3 System safety. Ensure the EA floor mat is free of any projection points or sharp edges and burrs which could snag, jam, or damage clothing and equipment, trip or injure the occupant or maintenance personnel, or jeopardize service operations (see 4.14.3).

3.14.4 Marking. The EA floor mat shall use any combination of labels, markings, or imprinting indicating safety precautions and warnings to include a clear indication of which side is up to indicate the walking surface (see 4.14.4).

3.14.5 Service operations. The floor mats shall not interfere with the operation of the vehicle or soldier movement within (see 4.14.5).

4 VERIFICATION

4.1 First article inspection and lot acceptance tests (LATs). The EA floor mat system shall be verified to meet the Section 3 requirements of this specification through one or more of the following verification method(s) in accordance with Table I. If any sample fails to comply with the specified performance requirements in Section 3, the sample and/or lot shall be rejected (see 3.1).

a. Examination: shall consist of activities using visual evidence or physical examination of the item for requirements verification.
b. Analysis: shall consist of technical or mathematical evaluations, mathematical models, simulations, algorithms, charts, diagrams, representative data, or other appropriate means to demonstrate compliance for requirements verification.
c. Demonstration: shall consist of proving, illustrating, or furnishing documents certifying compliance to relevant criteria to the standards or specifications referenced or other appropriate means to demonstrate compliance for requirements verification.
d. Test: shall consist of subjecting the item in a manner that verifies its performance under appropriate and specified conditions. Testing also includes the act of recording related measurements, and conducting analysis and evaluation of resulting data for requirements verification.

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TABLE I: Verification matrix.

METHOD OF VERIFICATION 1 – Examination 2 - Analysis 3 - Demonstration 4 - Test						
Requirement		Verify	Verification Methods			
Section	Description	Section	1	2	3	4
3.1	First article inspection	4.1	X	X	X	X
3.2	Materials	4.2			X	
3.3.1	Documentation	4.3.1	X			
3.3.2	Inspection conditions	4.3.2	X	X	X	X
3.3.3	Units of measure	4.3.3	X			
3.3.4	Different thicknesses	4.3.4	X	X	X	X
3.4.1	Test fixture and input	4.4.1	X		X	
3.4.2	Anthropometric test device (ATD)	4.4.2	X		X	
3.4.3	Further instrumentation	4.4.3	X		X	
3.4.4	Injury criteria	4.4.4	X		X	X
3.4.5	Test conditions	4.4.5	X		X	X
3.4.6	Test specimen size	4.4.6	X		X	X
3.5	Environmental considerations	4.5			X	
3.6	AIPH TEP toxicity clearance	4.6			X	
3.7	Wet surface traction	4.7				X
3.8	Self-draining water immersion	4.8	X	X		X
3.9	High storage temperature	4.9				X
3.10	Low storage temperature	4.10				X
3.11.1	Average heat release rate	4.11.1	X	X		X
3.11.2	Average peak heat release rate	4.11.2	X	X		X
3.12	Fungal susceptibility	4.12	X			X
3.13	Fluid contamination	4.13	X			X
3.14.1	Mounting	4.14.1		X	X	X
3.14.1.1	Adhesive bonding	4.14.1.1		X	X	X
3.14.1.2	Tools	4.14.1.2		X	X	X
3.14.2	Maintenance	4.14.2	X	X	X	
3.14.3	System safety	4.14.3	X	X	X	
3.14.4	Marking	4.14.4	X	X	X	
3.14.5	Service operations	4.14.5	X	X	X	

4.2 Materials. Verify the list of materials (see 3.2).

4.3 General.

4.3.1 Documentation. The PA shall examine documentation as necessary (see 3.3.1).

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4.3.2 Inspection conditions. Verify through analysis the environmental conditions (see 3.3.2).

4.3.3 Units of measure. Verify through demonstration (see 3.3.3).

4.3.4 Different thicknesses. Demonstrate all floor mats thicknesses meet the requirements specified in the contract (see 3.3.4).

4.4 Survivability.

4.4.1 Test fixture and input. The EA floor mat shall be evaluated and verified a minimum of 3 times by the GVSC Occupant Protection Lab (OPL) in accordance with Test Procedure LQP-25-04 (see 3.4.1).

(The specific impact velocity shall be stated in the vehicle specific contract.)
(The FITA is a test fixture to simulate an underbody blast event, consisting of a nitrogen-driven piston which accelerates a striker plate vertically against a floating plate which carries the test specimen.)Anthropomorphic test device (ATD). Test in accordance with LQP-25-WI07 ensuring the knee and pelvis joints are at an angle of 90 degrees and has boots (see 6.3) on (see 3.4.2).

4.4.3 Further instrumentation. Demonstrate high-speed video, accelerometers placement, and test items meet the required temperature referenced in requirement 4.4.5 in accordance with LQP-25-WI08 (see 3.4.3).

4.4.4 Injury criteria. Analyze the injury responses (see 3.4.4)

4.4.5 Test Conditions. Verify test conditions are in accordance with NHTSA 208-14 Section 12-2 for using the lower half ATD in floor mat testing. Unless otherwise specified in the contract, all testing regarding requirements utilizing the lower half ATD in section 3.4 “Survivability” shall be conducted under the following ambient environmental conditions after exposure of the lower half ATD for a period of not less than 4 hours (see 3.4.5):

Temperature:	20.6°C - 22.2°C (69°F - 72°F)
Barometric pressure:	facility ambient
Humidity:	10 - 70% relative humidity

4.4.6 Test specimen size. Demonstrate the test specimen size (see 3.4.6).

4.5 Environmental considerations. Demonstrate no toxic chemicals exceeding limits as defined in 29 CFR 1926.64 or ozone-depleting chemicals as defined by 40 CFR 82 as Class I and Class II were utilized in the construction of the EA floor mat (see 3.5).

4.6 AIPH TEP toxicity clearance. An examination of the toxicity clearance shall be performed. The toxicity clearance shall include both a clearance of the EA floor mat materials final chemical

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formulations and a clearance of the EA floor mat materials final chemical formulations pyrolysis and combustion products. The toxicity clearance shall be obtained on each specific EA floor mat application or use condition, and new uses or change in EA floor mat formulations shall obtain a new toxicity clearance (see 3.6).

4.7 Wet surface traction. Test the walking surface wet static coefficient of friction in accordance with ANSI/NFSI B101.1 utilizing a Neolite rubber pad with hardness 85 to 92, Type A in accordance with ASTM D2240 to assess the traction (see 3.7).

4.8 Self-draining water immersion. Measure the initial weight (W_i) prior to test. Test in accordance with MIL-STD-810 Method 512 Procedure I. Condition the sample per section 2.3.2.2.a.(2) for 30-minutes, followed by complete immersion for 30-minutes, followed by self-draining on a 5° slope for 5-minutes, followed by the test in section 4 using seven immersion cycles. Measure the final weight (W_f) within 15 minutes, after exposure. The percent water retention ($WR_{\%}$) is calculated by subtracting the initial weight from the final weight and dividing by the initial weight and multiplying by 100%: ($WR_{\%} = \frac{W_f - W_i}{W_i} \times 100\%$). Allow the after exposure sample to dry. If inspection of material determines the exposure may have hindered EA, test the after exposure sample in accordance with section 3.4.4. Verify mat is self-draining, the material does not retain more than 10% of its weight on, below and within it after water immersion, and exposure to water has no adverse material effects (see 3.8).

4.9 High storage temperature. Verify in accordance with MIL-STD-810 Method 501 Procedure I, the high storage temperature is 71 ±2°C (160 ±4°F) (see 3.9).

4.10 Low storage temperature. Verify in accordance with MIL-STD-810 Method 502 Procedure I, the low storage temperature shall be -46 ±2°C (-51 ±4°F) (see 3.10.

4.11 Heat release rate (fire resistance).

4.11.1 Average heat release rate. Test the materials used in the EA floor mat in accordance with ASTM E1354. Verify heat release in a 300-second period test using a heat flux source of 50 kW/m^2. Average (minimum 3 samples) heat release rates for each time interval from ignition, and to 60 sec, and to 180 sec, and to 300 sec shall be less than 60 kW/m^2. Verify there is no melting, dripping or pooling of material when tested (see 3.11.1).

4.11.2 Average peak heat release rate. Verify the average peak heat release rate with a minimum of 3 samples is less than 85 kW/m^2. Verify there is no melting, dripping or pooling of material(s) when tested (see 3.11.2).

4.12 Fungal susceptibility. Verify in accordance with MIL-STD-810 Method 508 and using the fungi listed in Method 508 Table I (see 3.12).

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4.13 Fluid contamination. Verify in accordance with MIL-STD-810 Method 504 Procedure I for the items listed in MIL-STD-810 Method 504, Table 504.2-I: Solvents & cleaning fluids, Insecticides, and Disinfectants (no specific order required) (see 3.13).

(This list is not exhaustive and may include other fluids specified in the contract.)

4.14 Integration verification.

4.14.1 Mounting. Verify in accordance with MIL-STD-1472 and the TDP (Technical Data Package) through demonstration, analysis, or the applicable tests specified in the contract (see 3.14.1).

4.14.1.1 Adhesive bonding. Verify through demonstration or the applicable tests specified in the contract (see 3.14.1.1).

4.14.1.2 Tools. Verify through demonstration or the applicable tests specified in the contract (see 3.14.1.2).

4.14.2 Maintenance. Verify in accordance with MIL-STD-1472 through demonstration, analysis or test (see 3.14.2).

4.14.3 System safety. Verify through inspection (see 3.14.3).

4.14.4 Marking. Verify through inspection has labels and/or markings that indicate safety precautions and warnings (see 3.14.4).

4.14.5 Service operations. Demonstrate the floor mats remain affixed, do not crumple and become a tripping hazard when subjected to conditions stated herein (see 3.14.5).

5 PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order. When packaging of materiel is to be performed by DoD or in- house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Specifications covered by this standard are intended for use in acquisition to obtain military-unique items. Non-Government standards or commercial item descriptions should be used to describe the requirements for a commercially available item.

6.2 Administrative requirements.

6.2.1 Quality documents supporting inspection, test, and analysis. When requested, all quality documents supporting inspection will be made available to the Government. At the time of any Government request, legible design or manufacturing drawings and specifications will be made available. Such drawings and specifications should be annotated to reflect the latest Government approved revision incorporated with an updated cross-reference parts list for major subassemblies and size/grade specifications for common hardware. Upon completion of inspection(s) by the Government, all drawings and specifications should be returned. The Government reserves the authority to inspect end items or any parts/components during all manufacturing processes and reject any items not conforming to drawings and/or specifications. All deficiencies detected during any end item or in process inspection should be rejected and replaced with an item conforming to the drawing and/or specifications at no cost to the Government before acceptance. During any Government inspection, the contractor should provide inspection assistance upon request.

6.2.2 Inspection and test equipment. The Contractor is responsible for ensuring all supplies and services conform to the contract requirements. If no other calibration specification and/or requirements are specified in this contract, then the inspection and test equipment should be calibrated in accordance with the suggested Contractor's guidelines traceable back to the National Institute of Science and Technology (NIST) standards.

6.2.3 Test and evaluation facilities. The Contractor should provide the government with a copy of test set-up procedures and test rig/equipment used to verify requirements in Table I, Verification of requirements, are met. The Government reserves the right to witness any or all of the inspections, demonstrations, tests and/or analysis.

6.2.4 Certifications. Where certification is required to verify material or component to contract requirements, the Contractor should furnish such certification along with applicable substantiating analytical data, documented test, and performance data. The certifications should be made available for Government review during Prototype Verification Inspection (PVI) / First Production Verification Inspection (FPVI), and copies of the certifications should be provided to the Government upon request. Certifications should include all documentation, examinations, and tests where applicable. Certification of verification to specific contract

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and/or specification requirements should be a statement signed by the Contractor's Program Manager, to the effect that the Contractor has complied.

6.2.5 Unacceptable certifications. If any certification is unacceptable to the Government, the Contractor should conduct additional examinations/tests to provide additional documentation as required to verify certification at no increase in contract price. The Government may conduct tests to verify certifications.

6.2.6 Process certifications. When a process certification is required, it should include a written description of the processes and all associated work instructions.

6.2.7 Material certifications. When a material certification is required, it should include a copy of the material analysis. If the material is purchased from a subcontractor, a copy of the purchase order is also required.

6.2.8 Test certifications. When a test certification is required, it should include the following information as substantiation:

a. Drawing Number (if applicable)
b. Specification Title, Number, Edition
c. Number of Specimens Tested, Requirements
d. Actual Results Obtained, and Purchase Orders for Subcontracted Products

6.2.9 Government verification. All verification may be accomplished by analysis, demonstration, examination, testing, or any combination thereof. The contractor operations performed should be subject to Government verification at unscheduled intervals. Surveillance by the Government may occur to determine if practices, methods, and procedures of the contractor's approved (by the Government Procurement Contracting Officer), written Quality Assurance System Plan are being properly applied. The Government reserves the right to witness any or all of the inspections, demonstrations, tests and/or analyses.

6.3 Definitions. When using this document, refer to this section for guidance on terminology.

a. Combat boot is used to reference the following boot designation: Desert, Hot Weather NSN series 8430-01-514-5157 equipped with the military specification compliant Vibram® three layer sole (Size 11R for 50th percentile ATD, 9.5R for 5th percentile ATD, and 12W for the 95th percentile ATD).
b. EA floor mat is the final chemical formulation containing: a walking surface (for conducting service operations and maintenance), energy attenuation materials (for protecting against injuries due to accelerative forces during blast events), coating materials (for performance), adhesive bonding materials (for construction), and all other materials (for requirements compatible characteristics) contained within the mat.

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c. GVSC lower body fixture identifies a test device used in LQP-25-WI09 with the following characteristics:
    1. The fixture should be capable of being mounted onto the FITA working platform such that it doesn't get in contact with the moving floating plate or the adjustment shafts.
    2. The fixture should be capable of supporting the weight of the lower portion of a 5th, 50th or 95th ATD. The lower portion should include all parts and instrumentation from the pelvis down in accordance with 49 CFR 572. The fixture should have the means to mount the ATD pelvis using the four (4) screws which are designed to hold the lumbar load cell.
    3. The fixture should be capable of shifting the ATD mount location from 16 to 24 inches from the platform, in 0.5 inch increments to allow for testing of floor mats of multiple thickness while maintaining a posture in which the hip, knee and ankle pivots are each fixed to 90°.
d. Hinder EA identifies an inspection technique for determining unacceptable immediate or potential long-term energy attenuation material characteristics including, but not limited to: degrading tibia axial compressive force performance or tibia bending moment performance due to obstructing, clogging, jamming, loosening, weakening, binding, sticking, or deforming from differential rates of expansion or contraction of materials.
e. No adverse material effects identifies an inspection technique for determining unacceptable immediate or potential long-term EA floor mat characteristics including, but not limited to: hindered EA performance; abrasion or erosion of walking surfaces; penetration, deformation, or rupture of seals; oxidation or galvanic corrosion; chemical reactions or incompatibility interactions; cracking, blistering, peeling, buckling, de-laminating or separation of plies; coating, adhesive or fiber failure; curling or twisting tendencies; melting, softening, sweating, or swelling; fading or discoloration; hardening or embrittlement; shortening of operating life; or conditions that create safety hazards, repair problems, or maintenance issues.
f. Objective requirements (O) are the desired performance levels that should be met.
g. Previously untested sample is used to reference an EA floor mat that has not been previously tested under the conditions of section 4.4.
h. Threshold requirements (T) are the minimum acceptable performance levels that should be met by the EA floor mat, below which performance fails. Requirements specified in section 3 are defined as threshold requirements.

6.4 Injury Criteria. Injury criteria for the 5th, 50th, and 95th percentile male can be evaluated in accordance with TR 27790. The OCP TECD Report TR-27790 "Enhanced Injury Assessment Reference Values" was created to meet the OCP TECD program challenge statement of "designing for and protecting the Soldier population". The OCP Enhanced IARVs (e-IARVs) have not been adopted by US Department of Defense (DoD) Live Fire Testing & Evaluation (LFT&E), Army Evaluation Command (AEC), nor the Office of the Director, Operational Test and Evaluation (DOT&E), as of 19-September-2016, for use in current and/or new acquisition programs. As described in the report, the injury criteria developed for the OCP TECD program

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utilized existing biomedical research for the 5th percentile female, 50th percentile male, and 95th percentile male. This report recommends that PEOs, PMs, and RDECs may integrate these e-IARVs or extract information into contracting documents as appropriate in conjunction with the currently accepted injury criteria. These values are to be utilized for informational and developmental purposes only.

6.5 Subject term (key word) listing.

Accelerative forces
ATD
Attenuating
Blast events
Energy
Fixture
Floor mat
IARV
Injury
Materials

6.6 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodian:
Army – AT
DLA - CC

Preparing Activity:
Army – AT

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Review Activity:
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